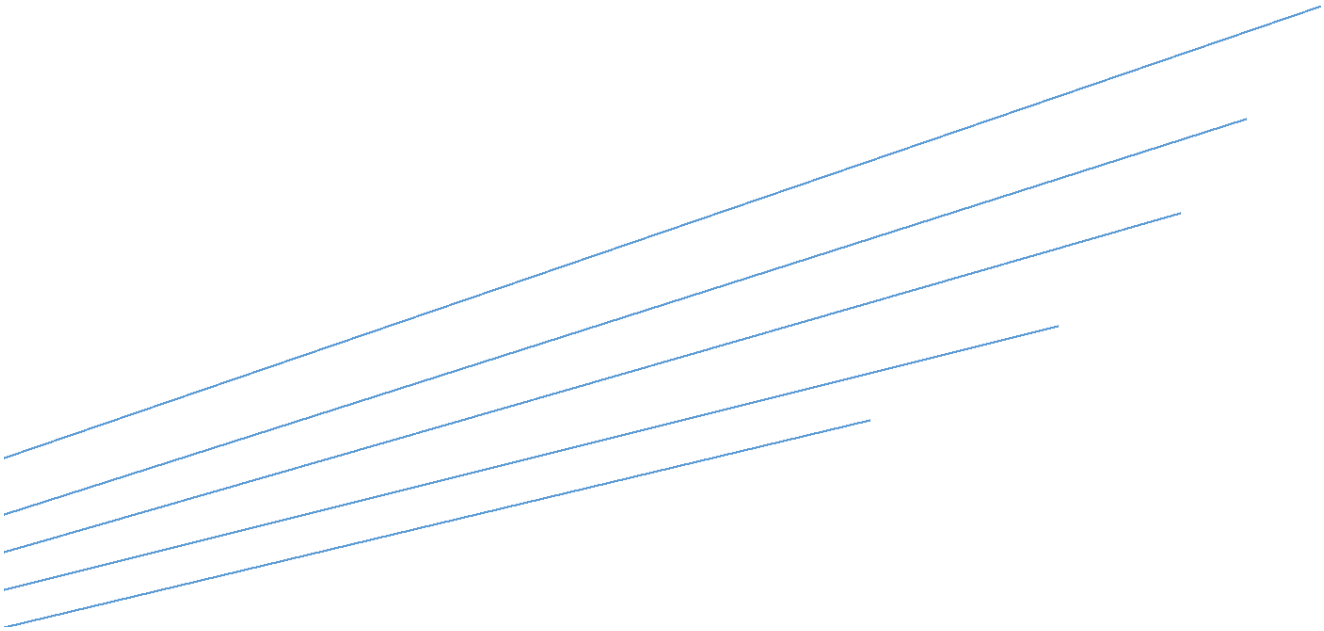
INTERNSHIP DAILY REPORT

Day 0 – Lab Environment Setup (Kali Linux Installation)

CyberShelter Cybersecurity Internship | Week 1



Mohamed Shameem PA

Associate – Information Security (Intern)

10 November 2026

1. Objective

The objective of Day 0 was to set up the lab environment for the internship by installing Oracle VirtualBox, creating a Kali Linux virtual machine with the required specifications, installing Kali Linux, and updating the system, so that all training tasks can be performed without any setup issues.

2. Lab Environment

- **Host Machine:** Windows PC (AMD Ryzen 5 3450U)
- **Virtualisation Software:** Oracle VirtualBox
- **Guest OS:** Kali Linux (64-bit, Debian-based)
- **Installation Media:** Kali Linux installer ISO

3. Topics Covered

3.1 Virtualisation

Virtualisation means running one or more virtual computers on a single physical computer. Each virtual computer works like a real one, with its own operating system, memory and storage, but it shares the hardware of the host machine.

3.2 Hypervisor

A hypervisor is the software that creates and runs virtual machines. It divides the host's CPU, RAM and storage between the virtual machines.

- **Type 1 (Bare-metal):** runs directly on the hardware, for example VMware ESXi and Microsoft Hyper-V.
- **Type 2 (Hosted):** runs on top of an existing operating system, for example **Oracle VirtualBox** and VMware Workstation.

3.3 Virtual Machine (VM)

A virtual machine is a computer that runs inside software. It is isolated from the host, so anything done inside it, like testing tools, does not affect the main system. This makes it safe for security practice.

3.4 ISO Image

An ISO file is a single file that contains a complete copy of an installation disc. It is used as a virtual DVD to boot and install an operating system inside the VM.

3.5 Kali Linux

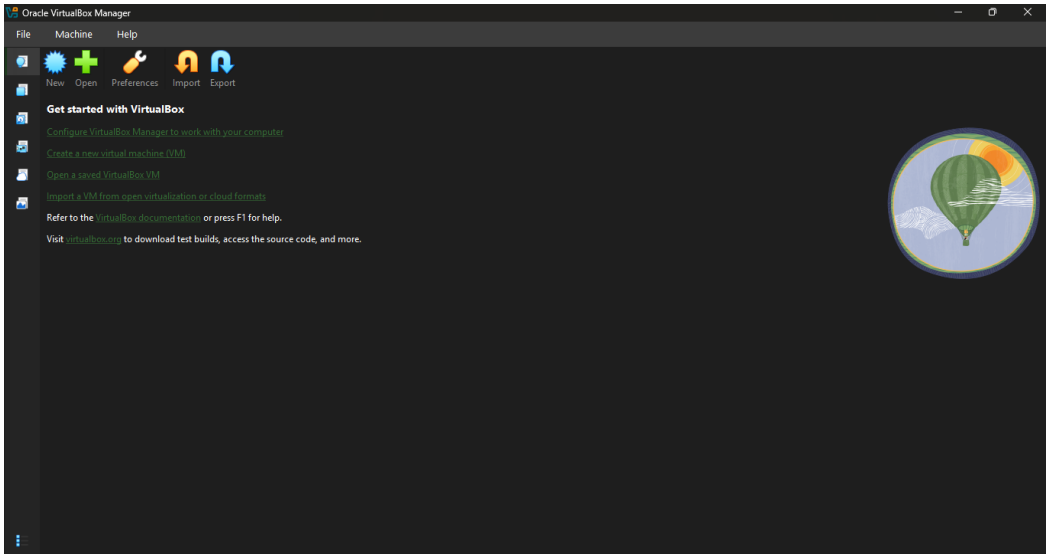
Kali Linux is a Debian-based Linux distribution designed for cybersecurity, penetration testing and digital forensics. It comes with hundreds of pre-installed security tools.

4. Tasks Performed

4.1 Create the Virtual Machine

4.1.1 Open VirtualBox

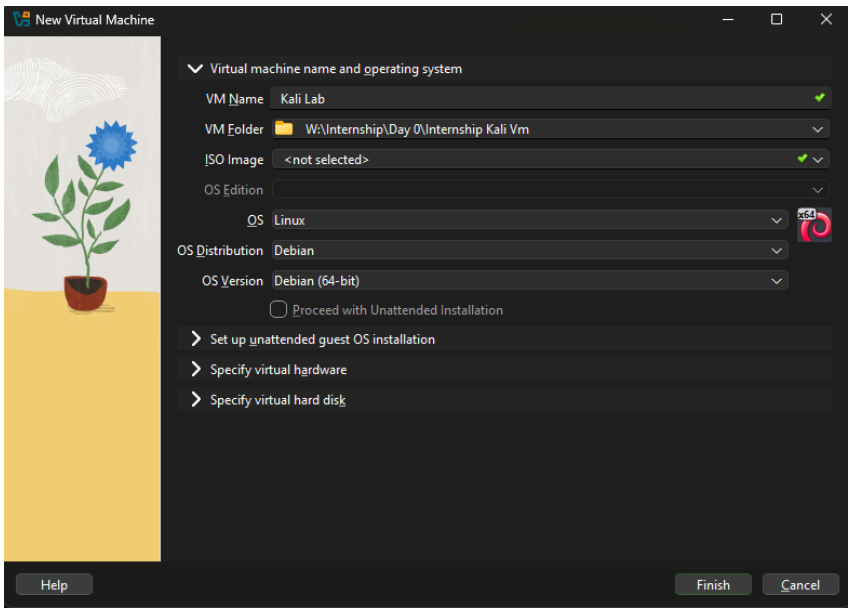
- **Purpose:** Start the virtualisation software to create a new VM.
- **Observation:** Oracle VirtualBox Manager was opened and the **New** option was selected to create a virtual machine.



VirtualBox Manager home screen.

4.1.2 VM Name and Operating System

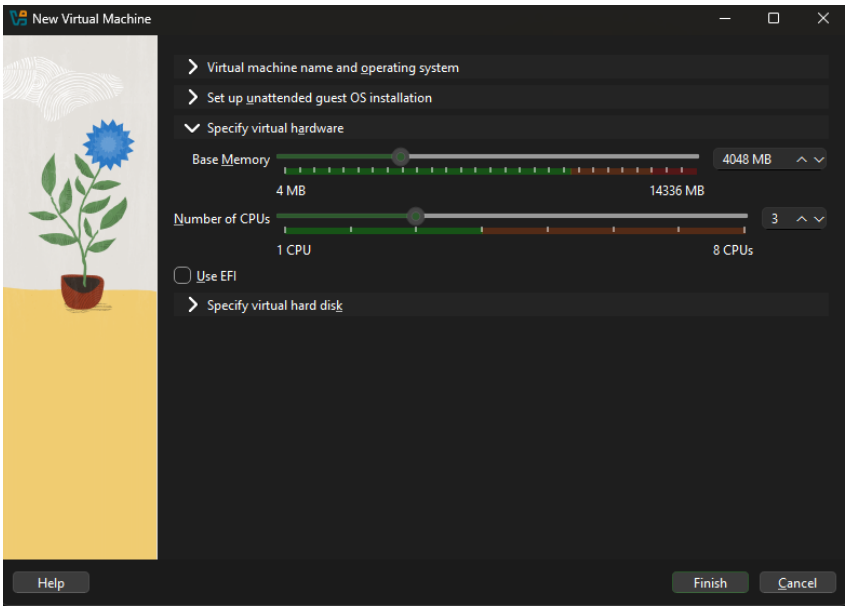
- **Purpose:** Set the VM name, storage location and OS type.
- **Observation:** The VM was named **Kali Lab**, with OS type **Linux**, distribution **Debian** and version **Debian (64-bit)**, since Kali Linux is based on Debian.



VM name and operating system settings.

4.1.3 Memory and CPU

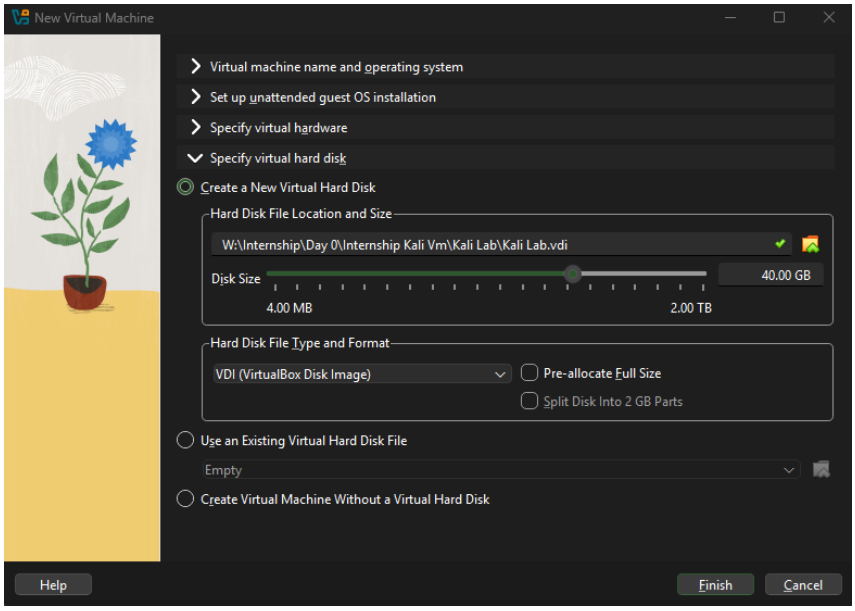
- **Purpose:** Allocate RAM and processor cores to the VM.
- **Observation:** Base memory was set to **4048 MB (about 4 GB)** and **3 CPUs** were allocated. EFI was left disabled, so the VM boots in BIOS mode.



Memory and CPU allocation.

4.1.4 Virtual Hard Disk

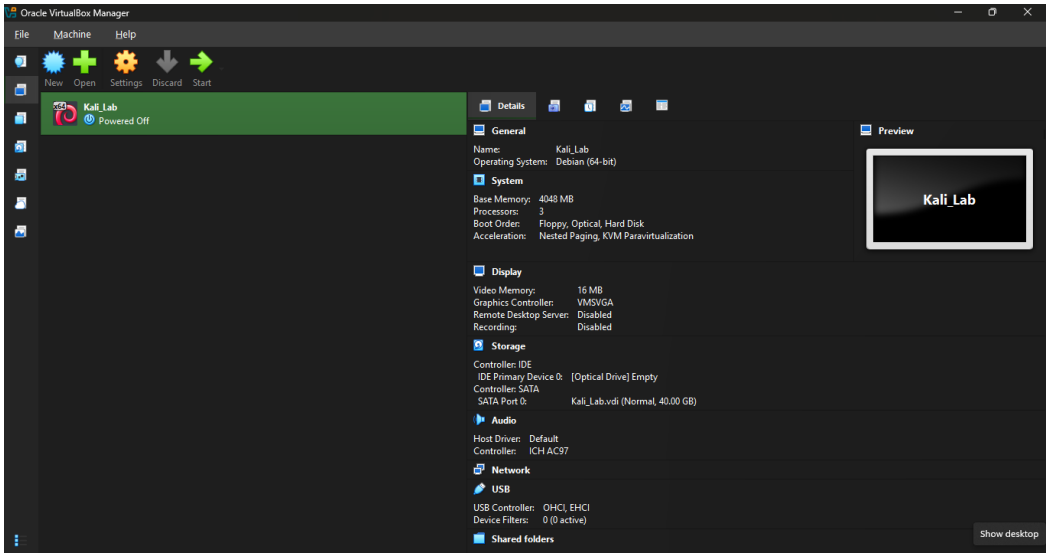
- **Purpose:** Create the virtual disk where Kali Linux will be installed.
- **Observation:** A new **40 GB VDI** (VirtualBox Disk Image) was created. It is dynamically allocated, so it only uses space on the host as data is added.



Virtual hard disk settings.

4.1.5 VM Created

- **Purpose:** Confirm the VM settings before starting.
- **Observation:** The **Kali Lab** VM was created and listed in VirtualBox with the selected settings.

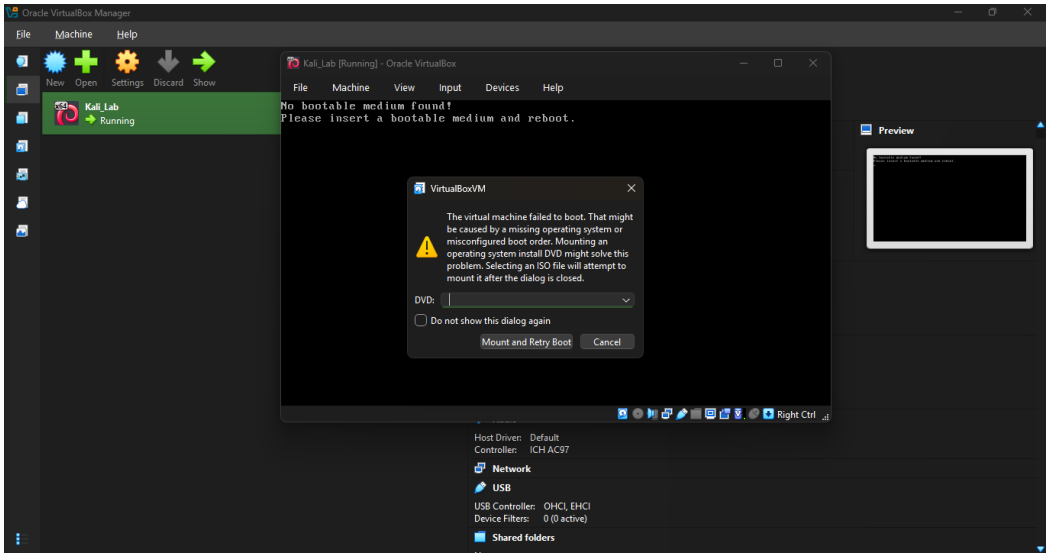


Kali Lab VM created in VirtualBox.

4.2 Boot the Installer

4.2.1 First Boot

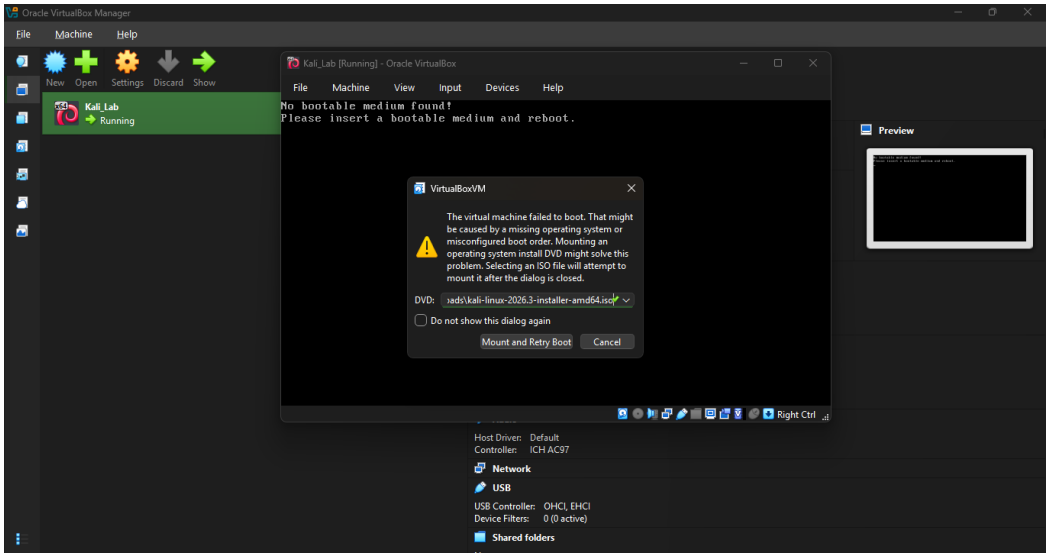
- **Purpose:** Start the VM for the first time.
- **Observation:** The VM showed a **"No bootable medium found"** message because no ISO was attached when the VM was created.



First boot without installation media.

4.2.2 Mount the Kali ISO

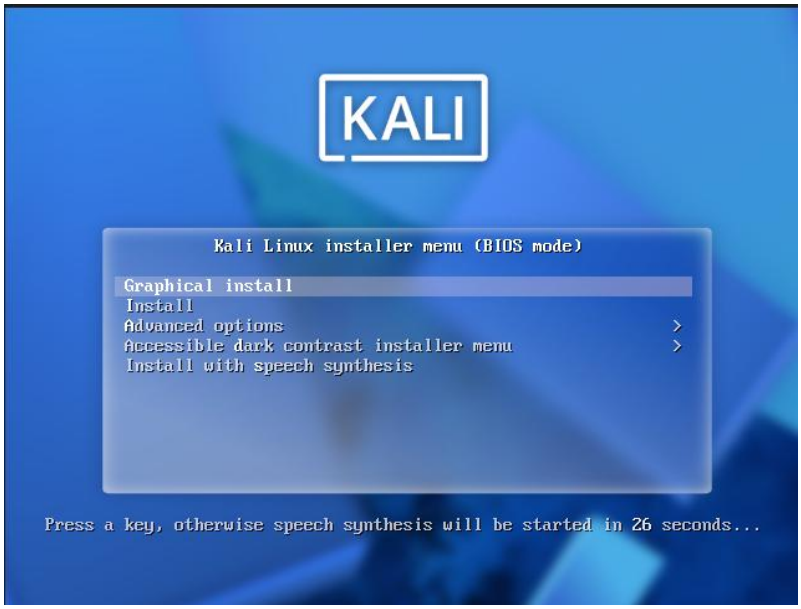
- **Purpose:** Attach the Kali Linux ISO as a virtual DVD.
- **Observation:** The Kali Linux ISO was attached to the VM's optical drive, and the VM was restarted.



Attaching the Kali ISO to the VM.

4.2.3 Kali Boot Menu

- **Purpose:** Choose the installation method.
- **Observation:** The Kali Linux installer menu (BIOS mode) appeared, and **Graphical install** was selected.

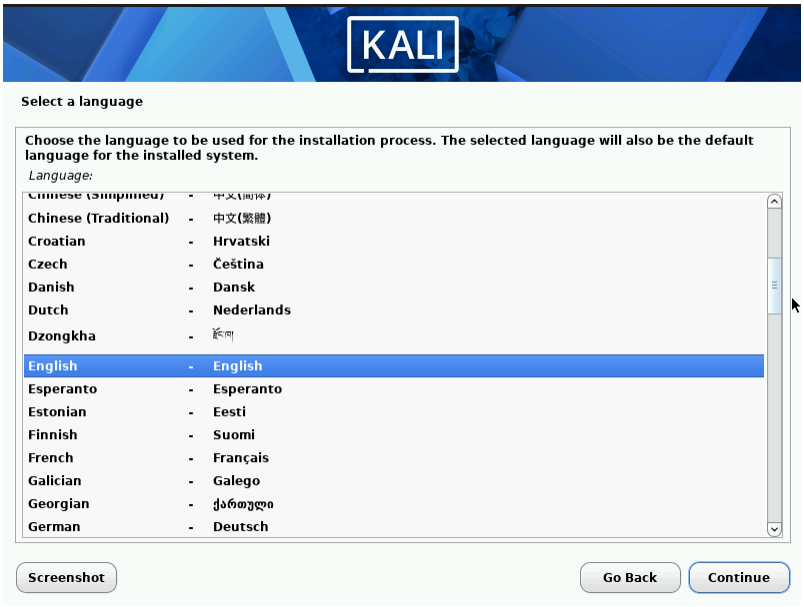


Kali Linux installer menu.

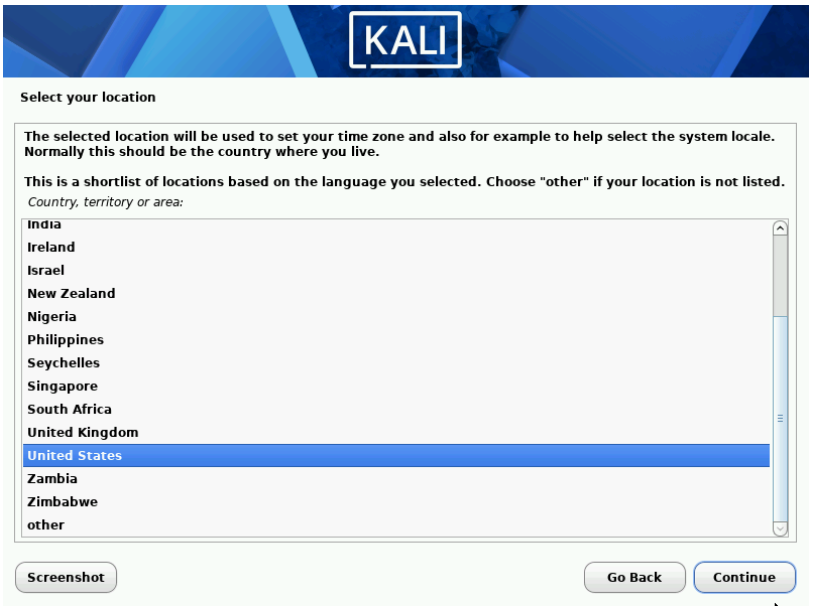
4.3 Install Kali Linux

4.3.1 Language and Location

- **Purpose:** Set the system language, location and time zone.
- **Observation:** English was selected as the language, followed by the location.



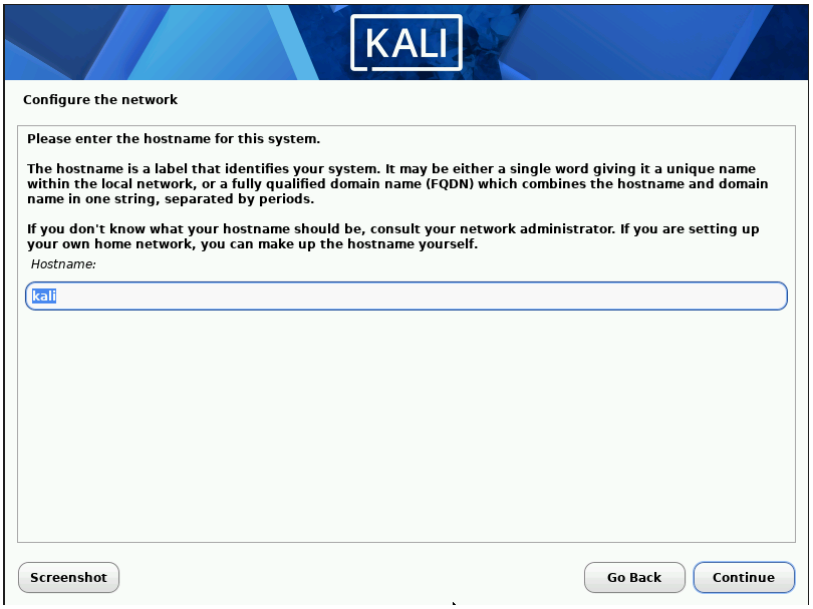
Selecting the installation language.



Selecting the location.

4.3.2 Hostname

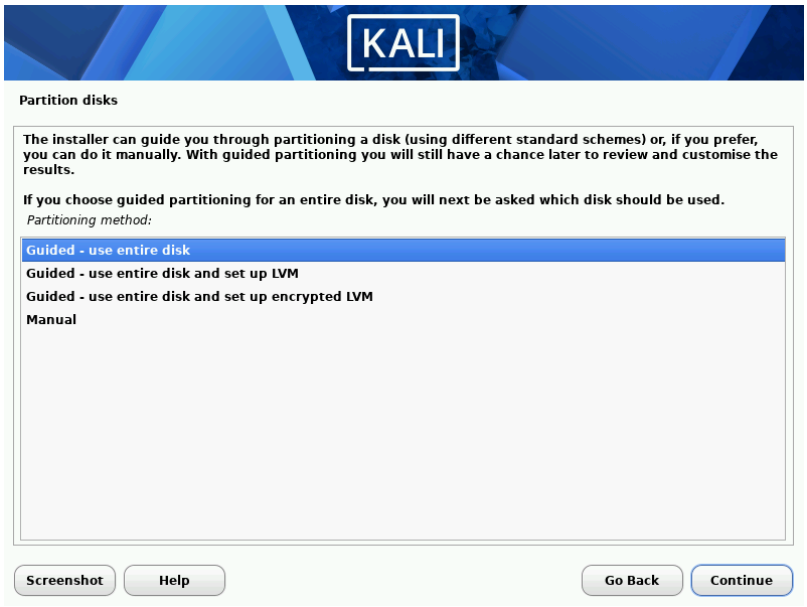
- **Purpose:** Give the system a name on the network.
- **Observation:** The hostname was set to **kali**.



Setting the hostname.

4.3.3 Disk Partitioning

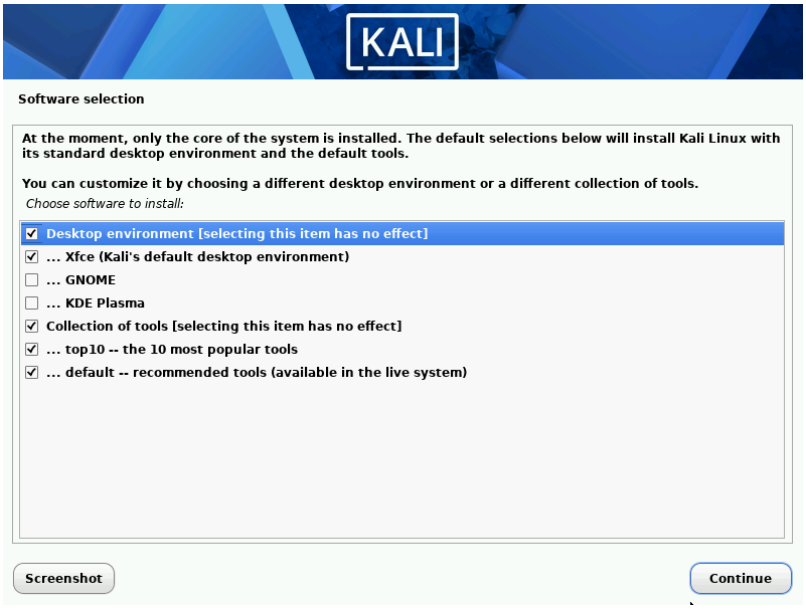
- **Purpose:** Prepare the virtual disk for installation.
- **Observation: Guided – use entire disk** was selected with all files in one partition. This created the root partition and a swap partition on the 40 GB disk.



Disk partitioning.

4.3.4 Software Selection

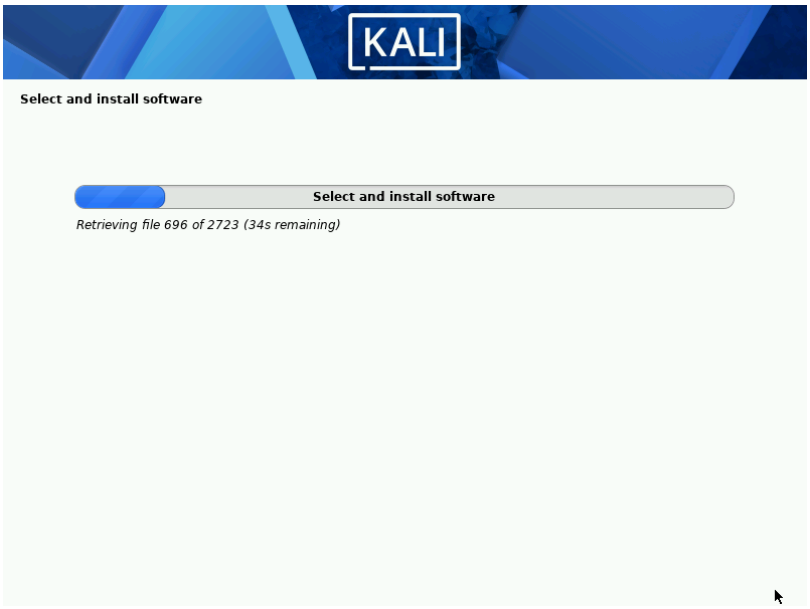
- **Purpose:** Choose the desktop environment and tool set.
- **Observation:** The default selection was kept: the **Xfce desktop** and the **top 10 Kali tools**.



Software selection.

4.3.5 Installing Software

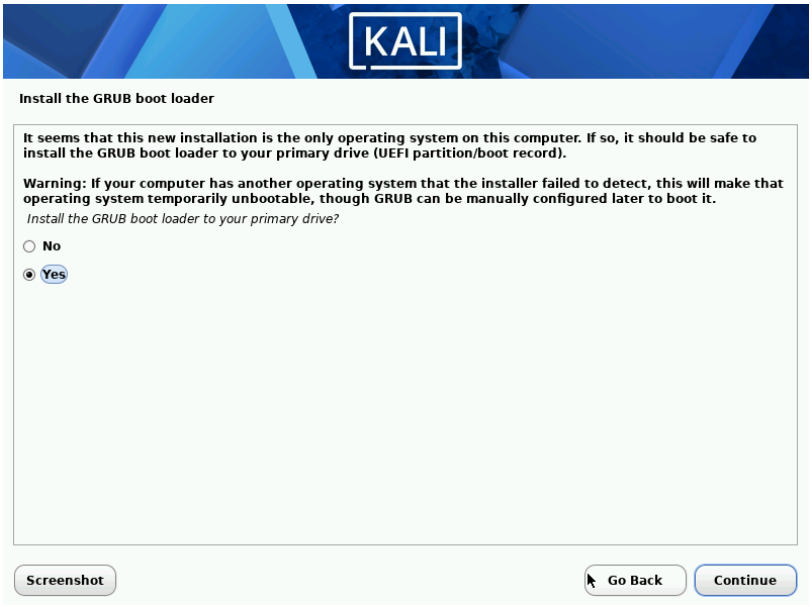
- **Purpose:** Copy and install the system files.
- **Observation:** The installer copied and installed the base system and the selected software.



Installation in progress.

4.3.6 GRUB Boot Loader

- **Purpose:** Install the boot loader so the system can start.
- **Observation:** The **GRUB boot loader** was installed on the primary disk (/dev/sda), which allows Kali Linux to boot.

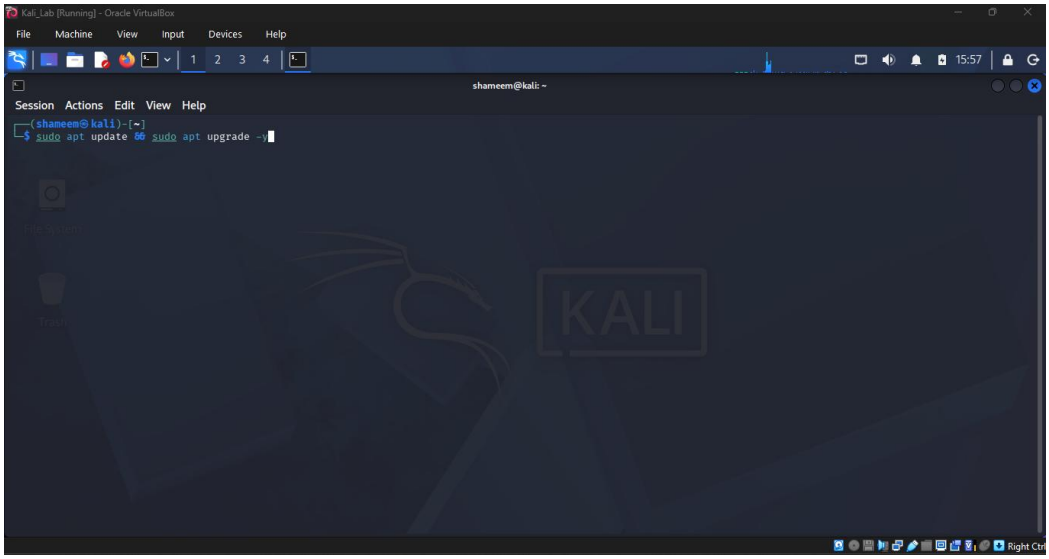


Installing the GRUB boot loader.

4.4 Update the System

4.4.1 System Update

- **Command Executed:** `sudo apt update && sudo apt upgrade -y`
- **Purpose:** Updates the package list and upgrades all installed packages to the latest versions.
- **Observation:** The package list was refreshed and all available updates were downloaded and installed successfully.



Updating Kali Linux.

5. Results

5.1 Final VM Configuration

Item	Configuration
VM Name	Kali Lab
Operating System	Kali Linux (Debian 64-bit)
RAM	4048 MB (approx. 4 GB)
CPU	3 cores
Storage	40 GB VDI (dynamically allocated)
Boot Mode	BIOS (EFI disabled)
Hostname	kali
Desktop	Xfce

5.2 Deliverable

The Kali Linux VM was installed successfully, **boots without any issues** and is fully updated, ready for the Week 1 tasks.

6. Key Learnings

- Understood what virtualisation and a hypervisor are, and the difference between Type 1 and Type 2 hypervisors.
- Learned how to create a virtual machine and allocate RAM, CPU and storage in VirtualBox.
- Learned the Kali Linux installation steps, including partitioning and installing the GRUB boot loader.
- Learned to keep the system up to date using `apt update` and `apt upgrade`.

7. Challenges Faced

- **No bootable medium on first boot:** The VM did not start the installer because no ISO was attached during VM creation. This was solved by attaching the Kali ISO to the VM and restarting the VM.

8. Conclusion & Next Steps

Day 0 successfully prepared the lab environment for the internship. A Kali Linux virtual machine was created in VirtualBox with 4 GB RAM, 3 CPU cores and 40 GB storage, installed and updated. The lab is now ready for the training tasks.

Next Steps: Day 1 will focus on computer fundamentals and exploring system hardware using Linux commands.